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Is the CS2 Skin Market Integrated with Traditional and Crypto Financial Markets? A Time Series Analysis

Time Series Analysis in Finance (TSA01_FS2601)

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Abstract

This paper examines whether the CS2 skin market — a multi-billion-dollar virtual goods market operating through the Steam Marketplace — is integrated with traditional and crypto financial markets. We construct a daily value-weighted price index from 1,607 liquid CS2 skins over the period August 2013 to June 2024 and test for market integration against the S&P 500, Bitcoin, and Gold using exploratory data analysis, GARCH(1,1) volatility modelling, and a VAR(4) model with Granger causality tests. We find no evidence of market integration: contemporaneous correlations between CS2 returns and all three comparison assets are indistinguishable from zero, and no Granger causal relationship is detected in either direction at the 5% significance level. CS2 exhibits extreme annualised volatility (114.56%) and GARCH persistence identical to Bitcoin (0.9990), but its price dynamics are driven by game-specific factors rather than macroeconomic forces. These results position CS2 skins as a genuinely segmented alternative asset class with potential diversification benefits for financial investors.

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1 Introduction

Counter-Strike 2 (CS2) is one of the most played competitive shooters in the world. Since 2013, players have been able to trade purely cosmetic weapon modifications — "skins" — through the Steam Marketplace at freely negotiated, real-money prices. Unlike the closed monetisation systems of competing titles, Steam operates an open secondary market, making CS2 skins an observable and quantifiable asset class. By early 2025, the ecosystem had grown to over 22,000 tradeable items with an estimated market value of $4.1 billion Guede-Fernández et al. (2025). Prior work has found attractive risk-adjusted returns and low correlation with traditional assets Dobrynskaya and Strelnikov (2025), but no study has formally tested market integration using a VAR and Granger causality framework on a comprehensively constructed price index.

This paper fills that gap. We construct a daily value-weighted CS2 price index from raw Steam Marketplace transaction data and ask:

"Is the CS2 skin market integrated with traditional and crypto financial markets, and what is the direction and temporal dynamics of any such relationship?"

We test this against the S&P 500, Bitcoin, and Gold — benchmarks for traditional equity, speculative crypto, and safe-haven performance — using EDA, GARCH(1,1) volatility modelling, and a VAR model with Granger causality tests.

2 Literature Review

The CS2 skin market constitutes a substantial financial asset class. Operating through the Steam Marketplace — an open secondary market with real-currency pricing — it reached an estimated $4.1 billion in value by early 2025 Guede-Fernández et al. (2025). Unlike competing titles that restrict trading to developer-set prices, CS2's open market allows supply and demand to determine prices, making skins suitable for standard financial analysis Zhou (2024). Dobrynskaya and Strelnikov document average annual returns of 41.2% with a Sharpe ratio of 0.34 across nearly 5,000 CS skins from 2013 to 2024 — exceeding both the US equity market (0.25) and gold (0.12) Dobrynskaya and Strelnikov (2025). These findings motivate the central question of whether the CS2 market operates independently of, or co-moves with, mainstream financial assets.

Bitcoin serves as the primary speculative comparator. Baur et al. find that Bitcoin is held predominantly for speculative purposes, with returns uncorrelated with traditional asset classes under both normal and stressed market conditions Baur et al. (2018). Bouri et al. confirm that Bitcoin offers minimal hedging properties, with its portfolio value lying in diversification rather than risk reduction Bouri et al. (2017). The shared demographic between Bitcoin investors and CS2 skin traders — young, digitally native, and risk-tolerant — motivates the hypothesis that crypto market sentiment may Granger-cause skin prices. Akin et al., using a VAR framework equivalent to the one applied here, find weak but statistically detectable interactions between Bitcoin, the S&P 500, and gold, establishing the empirical context against which CS2's behaviour is measured Akin et al. (2024).

The methodological toolkit draws on three established approaches. Vector Autoregression, introduced by Sims, captures joint dynamics across multiple time series without imposing a-priori causal structure Sims (1980). Granger causality Granger (1969) operationalises directional predictability: X Granger-causes Y if past values of X improve forecasts of

Y beyond Y's own history. GARCH(1,1) Bollerslev (1986) models time-varying conditional variance and its clustering properties, enabling cross-asset comparison of shock persistence.

3 Methodology

3.1 Hypotheses

Three testable hypotheses are derived from the literature:

H1 CS2 skin index returns are significantly correlated with Bitcoin returns (market integration hypothesis).

H2 Bitcoin Granger-causes CS2 skin prices, but not vice versa — motivated by the shared speculative, crypto-adjacent user demographic.

H3 CS2 exhibits higher volatility than the S&P 500 but similar volatility clustering patterns to Bitcoin.

3.2 Data

CS2 skin transaction data are sourced from a publicly available Kaggle dataset (CC-BY-NC-SA 4.0) containing Steam Marketplace records for 22,495 items from August 2013 to June 2024. The sample is restricted to wearable item categories (weapon skins, knives, gloves) and StatTrak variants are excluded. A liquidity filter drops any item recording fewer than two transactions in any calendar month SIX Swiss Exchange AG (2024), reducing the sample to 1,607 liquid items. The daily CS2 index level is:

$$I_t = \frac{\sum_i p_{it} \cdot v_{it}}{\sum_i v_{it}} \tag{1}$$

where p_{it} is the volume-weighted average price of item i on day t and v_{it} is its trading volume. This value-weighted aggregation mirrors market-capitalisation weighting in equity indices MSCI Inc. (2024), ensuring that heavily traded skins drive the index level. The index spans 3,960 trading days. Daily log returns for the S&P 500 (`^GSPC`), Bitcoin (`BTC-USD`), and Gold (`GC=F`) are retrieved via `quantmod`; all four series are aligned to a common sample of 2,450 observations.

3.3 Analytical Pipeline

Exploratory data analysis covers descriptive statistics, a contemporaneous correlation matrix, and Augmented Dickey-Fuller tests to confirm stationarity of log returns. A GARCH(1,1) model with student-t innovations is then fitted to each return series to compare unconditional volatility and shock persistence across assets (H3). Finally, a VAR(p) model is estimated on the joint system with the lag order selected by BIC; Granger causality is tested via F-tests on jointly excluded lag coefficients, covering both directions between each benchmark and CS2 (H1, H2).

4 Results and Discussion

4.1 Exploratory Data Analysis

Table 1 reports descriptive statistics for all four return series over the common sample of 2,450 observations. CS2 is the most volatile asset at 114.56% annualised, exceeding Bitcoin (69.87%), consistent with the first part of H3. The correlation matrix (Table 2) shows near-zero contemporaneous correlations between CS2 and all three benchmarks ($r_{\text{CS2,BTC}} = -0.016$, $r_{\text{CS2,SP500}} = 0.000$, $r_{\text{CS2,Gold}} = 0.011$), constituting preliminary evidence against H1. ADF tests reject the unit root null at the 1% level for all series; all subsequent analyses are performed on log returns without further differencing.

Table 1: Descriptive Statistics — Daily Log Returns

Asset	Mean (%)	Std Dev (%)	Skewness	Ex. Kurtosis	Ann. Vol. (%)
CS2	0.009	7.217	−0.37	3.82	114.56
S&P 500	0.041	1.133	−0.80	15.74	17.98
Bitcoin	0.203	4.401	−0.63	9.07	69.87
Gold	0.026	0.927	−0.06	3.77	14.71

Table 2: Correlation Matrix — Daily Log Returns

	CS2	S&P 500	Bitcoin	Gold
CS2	1.000	0.000	−0.016	0.011
S&P 500	0.000	1.000	0.213	0.020
Bitcoin	−0.016	0.213	1.000	0.084
Gold	0.011	0.020	0.084	1.000

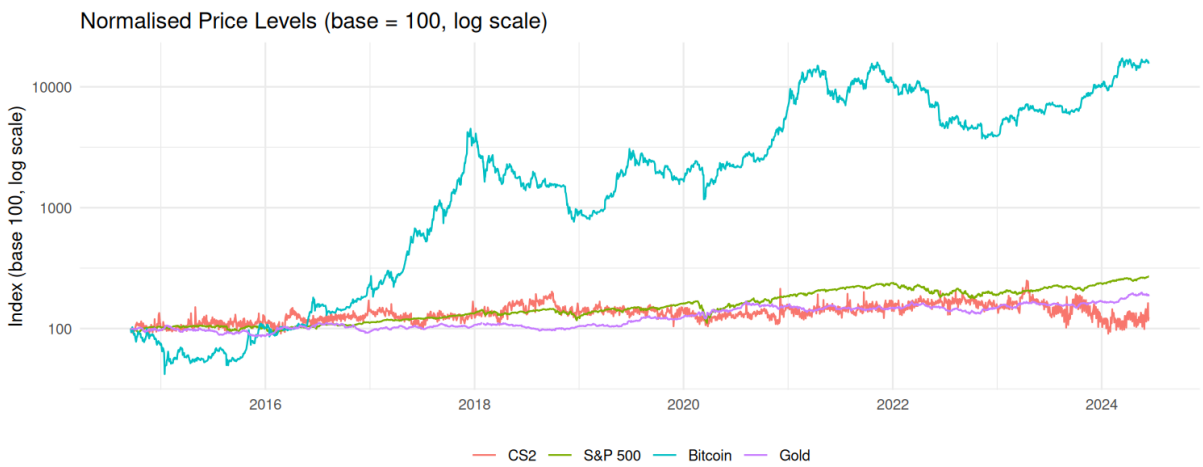

Figure 1: Normalised price levels — CS2 index, S&P 500, Bitcoin, and Gold (2013–2024, rebased to 100 at start of common sample; log scale).

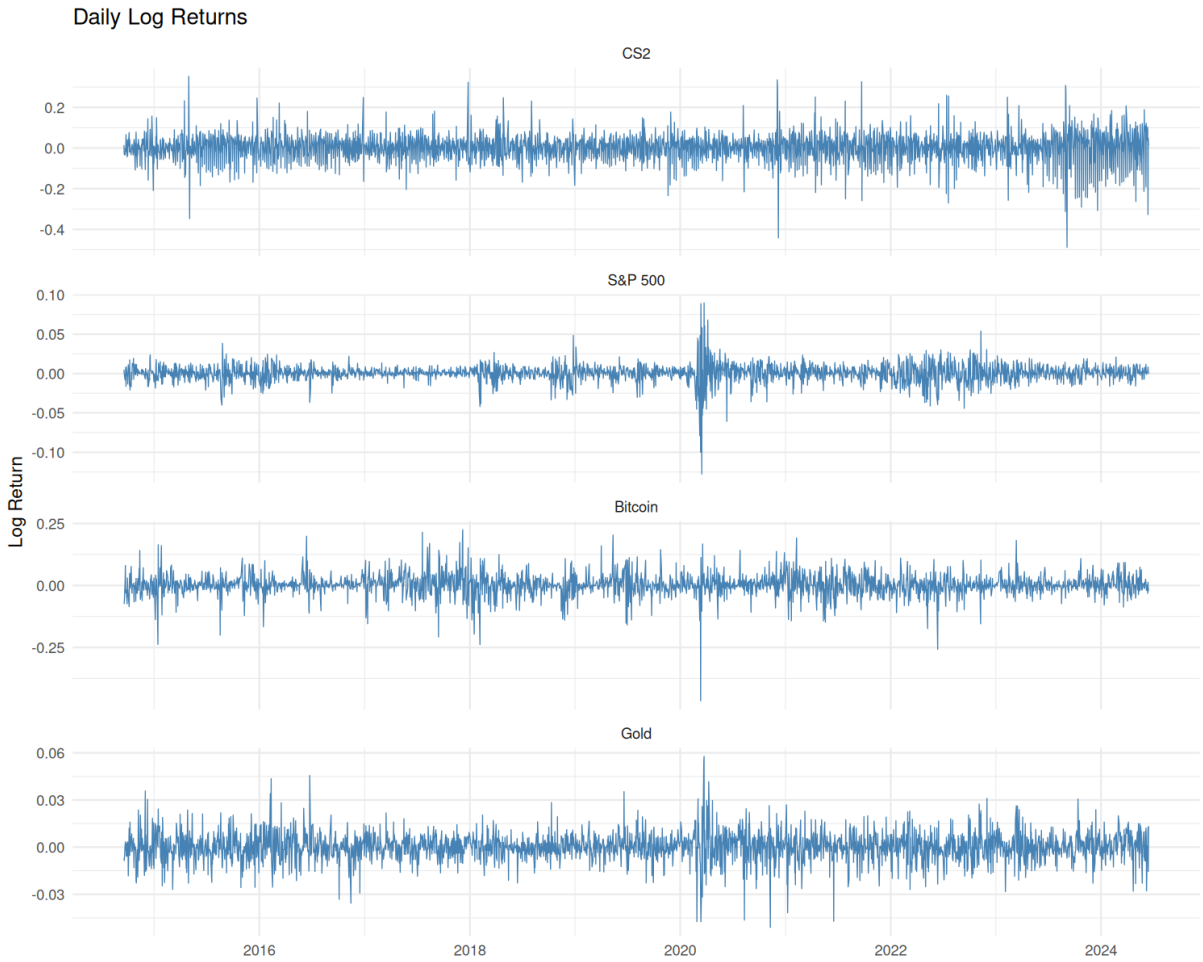

Figure 2: Daily log returns — CS2 index, S&P 500, Bitcoin, and Gold (2013–2024). CS2 and Bitcoin exhibit substantially wider return swings than the S&P 500 and Gold throughout the sample.

4.2 GARCH Volatility Comparison (H3)

Table 3 reports estimated GARCH(1,1) parameters. H3 is *partially supported*. CS2 unconditional volatility (161.88%) more than triples the S&P 500 (49.70%), confirming the first part. The persistence parameter $\hat{\alpha} + \hat{\beta} = 0.9990$ is identical for CS2 and Bitcoin, supporting clustering similarity. However, CS2's ARCH coefficient (0.0151) is far lower than Bitcoin's (0.1265): CS2 volatility does not spike sharply after individual shocks but persists in long-duration regimes, consistent with the Steam Marketplace's slow price-adjustment microstructure.

Table 3: GARCH(1,1) Estimates

Asset	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	Uncond. Vol. (%)
CS2	0.0151	0.9839	0.9990	161.88
S&P 500	0.1821	0.8158	0.9979	49.70
Bitcoin	0.1265	0.8725	0.9990	337.76
Gold	0.0280	0.9642	0.9923	16.61

4.3 VAR and Granger Causality (H1, H2)

A VAR(4) model is estimated on the joint system (BIC criterion; AIC suggests $p = 9$). Table 4 reports Granger causality F-tests; each row tests whether the cause variable Granger-causes the *entire remaining system jointly.*

Table 4: Granger Causality Tests — VAR(4)

Cause variable	F-statistic	p-value	Conclusion
Bitcoin	1.453	0.134	Not significant
S&P 500	1.275	0.226	Not significant
Gold	1.353	0.181	Not significant
CS2	1.663	0.068	Not significant (borderline)

Both H1 and H2 are **rejected**. None of the three benchmarks Granger-causes the system, and the EDA correlations confirm zero contemporaneous co-movement. The borderline CS2 signal ($p = 0.068$) runs in the opposite direction to H2's prediction, suggesting that any detectable influence flows *from* CS2 rather than towards it. Together, these results position CS2 as a genuinely segmented market whose price dynamics are driven by game-specific factors — patch updates, tournament cycles, case releases — rather than by macroeconomic or crypto forces.

4.4 Critical Discussion

The Steam Marketplace's 15% transaction fee creates persistent arbitrage barriers that partly explain the low co-movement: short-selling skins is impossible and round-trip costs are prohibitive. The value-weighted index concentrates on high-volume skins, suppressing mean returns relative to an equal-weighted benchmark and accounting for the low mean return (0.009%/day) versus Dobrynskaya and Strelnikov's 41% annual figure Dobrynskaya and Strelnikov (2025). The September 2023 CS2 engine update may have introduced a structural break not captured by the constant-parameter models. Finally, the joint Granger specification is conservative; a pairwise CS2–Bitcoin VAR would provide a more focused robustness check.

5 Conclusion

This paper investigated whether the CS2 skin market is integrated with traditional and crypto financial markets. Using a daily value-weighted price index from 1,607 liquid skins over the period August 2013 to June 2024, we tested for integration against the S&P 500, Bitcoin, and Gold via EDA, GARCH(1,1), and a VAR(4) with Granger causality tests.

The answer is clear: the CS2 skin market is not integrated with any of the three comparison assets. Contemporaneous correlations are indistinguishable from zero and no Granger causal relationship is detected at the 5% significance level in either direction. H1 and H2 are rejected; H3 is partially supported — CS2 shares Bitcoin's volatility persistence ($\hat{\alpha} + \hat{\beta} = 0.9990$) but differs markedly in shock reactivity. The near-zero correlation with all mainstream assets implies meaningful diversification potential for investors already holding equity or crypto positions Dobrynskaya and Strelnikov (2025), but extreme annualised volatility (114.56%) and the absence of price-discovery linkages to

broader markets make CS2 skins a high-risk, idiosyncratic asset unsuitable as a standalone hedge.

The main limitations are the restriction to Steam Marketplace data (excluding third-party platforms such as Skinport or CS.Money), the value-weighting bias towards liquid skins, and the potential structural break at the CS:GO-to-CS2 engine transition in September 2023. Future work should test for this breakpoint explicitly, estimate a pairwise CS2–Bitcoin VAR as a robustness check, and incorporate data from third-party platforms and equal-weighted benchmarks to obtain a more complete picture of CS2 skin price dynamics.

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A Appendix

A.1 Full EDA Plots

A.2 Index Construction Details

A.3 GARCH Robustness

A.4 VAR Diagnostics

A.5 R Code

Listing 1: Index Construction (excerpt)

```
# TODO: Add key R code excerpts here
library(data.table)
library(arrow)
```